

SIDDHI PRAVIN LIPARE

MS by Research · CVIT, IIIT Hyderabad

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EDUCATION

MS by Research, Computer Science & Engineering	Jan 2026 – Present
<i>International Institute of Information Technology, Hyderabad</i>	CGPA 8.00
BS Data Science & Engineering	Dec 2021 – May 2025
<i>Indian Institute of Science Education & Research, Bhopal</i>	CGPA 8.41
Senior Secondary (Class XII), CBSE Board · 90.0%	2021
Secondary (Class X), CBSE Board · 95.0%	2019

EXPERIENCE

Centre for Visual Information & Technology (CVIT), IIIT Hyderabad	May 2025 – Dec 2025
<i>Applied Research Fellow</i>	Project Page
- Designed RoadTones – a dataset–model–evaluation framework for tone-controllable road-video captioning. Work presented at CVPR Findings and WiCV@CVPR Workshop. - Developed key components of the dataset generation pipeline and evaluation metrics, enabling comprehensive benchmarking against 10 state-of-the-art Video Large Language Models (Video LLMs).	
Indian Institute of Technology Madras	May 2024 – Jul. 2024
<i>Research Intern</i>	Certificate , Report
- Explored Super-Resolution techniques for 3D medical imaging, enhancing z-axis resolution in CT and MRI scans. - Utilized GANs, Diffusion Models, and NeRF to improve image clarity for better diagnostics.	

PROJECTS

vLLM Inferencing for Road Videos	Dec. 2025 – Jan 2026
<i>CVIT, IIIT-Hyderabad</i>	
Investigated and optimized the vLLM inference pipeline for the project demo of RoadTones, minimizing video quality degradation during inference to preserve caption generation performance.	
BS Thesis – Image Tampering Detector	Jan. 2025 – April 2025
<i>IISER Bhopal</i>	Report
Leveraged the use of Discrete Wavelet Transform along with an Adaptive Sub-Band Attention Module to capture frequency-spatial cues to enable precise localization of tampered regions in an image. Poster presented at CV4Science@CVPR Workshop 2025, and extended abstract accepted at WiML@ICML Workshop 2026.	
Improving Monocular Depth Estimation of Outdoor Images in Dim Light Conditions	Sept. 2024 – Nov. 2024
<i>IISER Bhopal</i>	Report
Integrated denoising and deblurring filters into the transformer-based DepthAnything V2 model to improve depth estimation in blurry and noisy nighttime images.	
Prediction of Mortality Rate of Heart Failure Patients Admitted to ICU	Sept. 2023 – Nov. 2023
<i>IISER Bhopal</i>	GitHub
A binary classification problem executed using supervised machine-learning techniques. Several classifiers and pre-processing methods were applied to determine which classifier gave the best results.	

SKILLS & HOBBIES

Languages	Python, C/C++
Technologies	PyTorch, TensorFlow, Keras, OpenCV, Huggingface
Hobbies	Dance (Bharatnatyam), Performing Arts (Acting), Table Tennis

ACHIEVEMENTS

CVPR Travel Grant & WiCV@CVPR Travel Grant — Selected to receive travel grants for attending CVPR 2026.	2026
ISRO Robotics Challenge — Among the Top 30 teams out of 400 in India.	2024
IEEE CAS Student Design Competition — 3rd place in the Asia-Pacific Region, Patent filed on 18/10/2023.	2022
IISER Aptitude Test — Secured AIR 1153	2021
Mathex Exam (Ganit Pradnya) — Among the Top 100 in Maharashtra State.	2017